Problem 1 ... an unfortunate trolleybus

3 point

A trolleybus of total mass $M_\mathrm{t} = 15\,\mathrm{tonnes}$ (without any passengers) and of volume $V_\mathrm{t} = 90\,\mathrm{m}$
s entering a bridge over a river. Unfortunately, the bridge is being reconstructed, and careles
workers forgot to put up no entry signs. The trolleybus falls into the river. What is the maximun
percentage of the trolleybus volume that can be occupied by passengers such that after the fal
he trolleybus ends up floating to the surface? Consider the density of the human body to b
he same as the density of water. The trolleybus is airtight.

Verča heard that there was a demand for a trolleybus problem

Problem 2 ... our antique clock

3 point

An antique pendulum clock has to be wound up every Sunday at the same time to show th correct time during the whole week. While winding up, one has to raise a weight $m = 5.6\,\mathrm{kg}$ *b* $= 31\,\mathrm{cm}$ *to enable the clockwork to drive the pendulum. The pendulum is of length* $l = 64\,\mathrm{cm}$ *and almost all its mass is located at the lower end. How much energy (on average) is dissipate during one swing?*

Jarda is late sometimes

Problem 3 ... flywheel

3 point

Let us assume that we have a homogeneous disc (flywheel) of a mass $M = 10m$ and radius R There is a fly called Ľubo of a mass m sitting on the flywheel at a distance $0.75R$ from th enter of the disc, while the disc (together with Ľubo) is rotating with an angular frequenc $\omega_0 = 1.20\,\mathrm{rad{\cdot}s^{-1}}$. Ľubo's friend Slavo, who has the same mass as Ľubo, suddenly sits at th dge of the disc, while he had zero angular momentum with respect to the axis of rotation o he disc. Assume that both flies are point masses and that the system (the flywheel + flies) i ot subjected to any external torques. What is the angular frequency of the disc after Slav its on it?

Flies had been flying around Tomáš's head

Problem 4 ... roofing

3 point

Martin is standing on a roof. The coefficient of static friction between his shoes and roof tile
s 0.7. *To what percentage of the initial value does the effective coefficient of friction betwee*
Martin and the roof decrease when Martin sits down, if doing so transfers 60 % *of his weigh*
rom shoes to his trousers? The coefficient of static friction between his trousers and roof tile
s 0.4.

Martin was learning to be a roofer

Problem 5 ... spring in an airplane

3 point

An ideal spring oscillator is placed in a climbing airplane that has a load factor 3g. How does th frequency of its oscillations change? Express your answer as the ratio of the original frequenc to the new one.

Vojta was oscillating in an airplane

Problem 6 ... focusing light

4 point

We have an aquarium and a thin converging lens with a small diameter, which has an optica power of $\varphi = 4.20\,\mathrm{D}$ in the air. We place a point light source on the optical axis of the lens a great distance from the lens. The lens is placed $d = 4.20\,\mathrm{cm}$ from the aquarium in such a wa that its optical axis is perpendicular to the wall of the aquarium. The aquarium is large enough ts walls are made of thin glass, and it is full of water. What is the distance from the center o he lens at which the rays coming from the source are focused?

Karel was thinking about optics

Problem 7 ... induced voltage

3 point

Consider a region with homogeneous magnetic field. The region has a rectangular cross-sectio with sides $a = 3.0\,\mathrm{m}$, $b = 2.0\,\mathrm{m}$ *and the magnetic induction vector* $B = 1.0 \cdot 10^{-3}\,\mathrm{T}$ *is perpen dicular to this cross-section. We place a straight wire in parallel to the side* a*, such that its fre ends lie outside the region with the magnetic field. We connect these ends to a voltmeter.*

We then move the wire in uniform linear motion at velocity $v = 0.20\,\mathrm{m{\cdot}s^{-1}}$ *along the side b i.e. in a direction perpendicular to the magnetic induction vector and to the side* a*. Wha voltage does the voltmeter show when the wire passes through the magnetic field?*

Jindra came up with a problem longer than its solution

Problem 8 ... a constant atmosphere

3 point

What would be the thickness of the Earth's atmosphere if it had constant density $\rho = 1.29\,\mathrm{kg{\cdot}m}$ *everywhere? Assume that the Earth is round and that the atmosphere has a mass of* $m = 5.157 \cdot 10^{18}\,\mathrm{kg}$.

Karel is always speculating

Problem 9 ... molar mumble

4 point

An enthomologist caught a "mol" and he wants to dry it. Calculate its mass after evaporatio of all the water, if you know that a particular ball of one mole of living "mols"

- *contains* $n = 3.346 \cdot 10^{19}\,\mathrm{mol}$ *of water molecules,*
- *has surface area equal to twenty four times the area of Moldova (including Transnistria)*
- *has density* $\rho = 27.4\,\mathrm{kg{\cdot}m^{-3}}$.

Translator's note: "mol" is a Czech word for a clothing moth.

Vojta was admiring Vivaldi's Summer's tremolos

Problem 10 ... don't lean out of the windows

4 point

A dignitary is taking a fiacre ride. While moving at a speed $v_\mathrm{k} = 2.0\,\mathrm{m{\cdot}s^{-1}}$*, he's eating an apple. When finished eating, he throws the apple core out of the window such that it gains horizontal velocity* $v_\mathrm{h} = 3.0\,\mathrm{m{\cdot}s^{-1}}$ *perpendicular to the direction of travel. The core then proceeds to hit an unsuspecting peasant in the head just when he is* $d = 1.0\,\mathrm{m}$ *away from the dignitary and is not moving. At what speed did the apple core hit the peasant's head? Consider the core to be a point mass.*

Lego often travels by train

Problem 11 ... a healthy mind in a healthy body

4 point

Let a flexible rope be fastened to a wall at one end. Danka is pulling the other end She is standing on a small rug that lies freely on the floor. What is the maximum distance measured from the wall) at which Danka can be pulling the rope, such that the elasticity of the rope does not drag Danka back towards the wall? The coefficient of static friction between the rug and the floor is $f = 0.45$*, while it is much larger between Danka's feet and the rug Danka weighs* $m = 55\,\mathrm{kg}$ *and the mass of the rug can be considered negligible. Assume that the rope is being pulled horizontally, its free length is* $l_0 = 1.97\,\mathrm{m}$ *and its stiffness is* $k = 164\,\mathrm{N{\cdot}m^{-1}}$

Danka was exercising

Problem 12 ... insidiously smoky

4 point

A typical ionization smoke detector contains a sample of $^{241}\mathrm{Am}$ *that corresponds to a radioac ivity of* $1\,\mu\mathrm{Ci}$*. How many such detectors at the minimum do we need to dismantle to ge a sufficient mass of* $^{241}\mathrm{Am}$ *to start a nuclear chain reaction? The critical mass of* $^{241}\mathrm{Am}$ *corre ponds to* $60\,\mathrm{kg}$*.*

Pepa always steals problems from textbooks

Problem 13 ... an electron source

4 point

Consider a point source that emits electrons of speed $v = 20\,000\,\mathrm{km{\cdot}s^{-1}}$ *in the plane* yz*, and i*
placed in a homogeneous magnetic field which has magnitude B *and is pointing in the directio*
of the x *axis. The electron source is surrounded by a tube of radius* $R = 10\,\mathrm{cm}$*. The axi*
of symmetry of the tube passes through the electron source and is parallel to the y *axis. Wha*
s the minimum magnitude of the magnetic field B *which prevents the emitted electrons fro*
ouching the tube? *Kiko has been reading famous "green lecture notes*

Problem 14 ... is it cold outside? 4 point

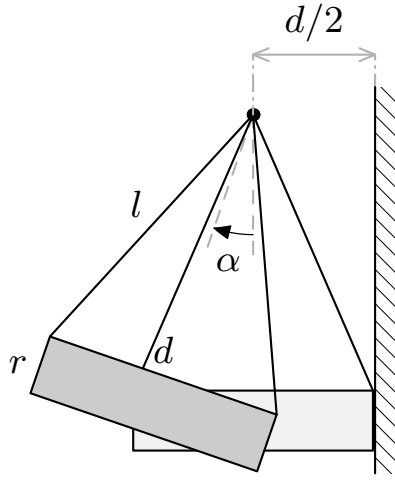
We are designing a thermometer. We want its range to be from $-40\,^{\circ}\mathrm{C}$ *up to* $+50\,^{\circ}\mathrm{C}$ *with the scale lines at every* $1\,^{\circ}\mathrm{C}$*. To distinguish the values on the thermometer properl just by eye from a distance of* $2\,\mathrm{m}$*, we need the angular distances between adjacent line to be at least* $4'$*. The radius of the capillary tube of the thermometer is* $0.2\,\mathrm{mm}$*. What i the minimum volume of alcohol that the thermometer needs to contain if it has to be able t show the full range of temperatures? We use alcohol with the volumetric thermal expansio coefficient* $\beta = 1.1 \cdot 10^{-3}\,\mathrm{K}^{-1}$*.* *Jarda will sell you a thermometer one day*

Problem 15 ... battering ram

5 point

We decided to conquer the Prague Castle. We will destroy the front gate with a battering ram in the shape of a thick homogeneous cylinder of length $d = 2.0\,\mathrm{m}$ and radius $r = 25\,\mathrm{cm}$, with its axis placed horizontally. We attach cables with lengths $l = 2.0\,\mathrm{m}$ to the upper points of both bases of the cylinder and fasten the other ends to a common point right above the center of the cylinder. With the ram assembled in this way, we come to the gate and place the ram in such a way that one of its bases touches the gate. Then we incline the ram $\alpha = 20^\circ$ around the hanging point and finally release it. At what speed will it hit the gate?

Jarda wants to be the president of FYKOS.



Problem 16 ... night-time thermodynamics lesson

4 point

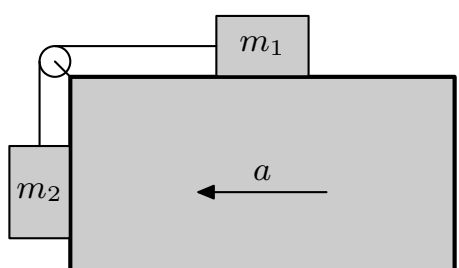
Verča got lost on a trip and decided to stay in an old shack overnight. The outside temperatur dropped to $t_0 = 10.0\,^\circ\mathrm{C}$ *at night, but Verča kept the inside temperature at* $t_1 = 16.0\,^\circ\mathrm{C}$ *However, the shack had a rectangular hole in the wall of dimensions* $a \times b$*, where* $a = 0.50\,\mathrm{m}$ *and* $b = 0.30\,\mathrm{m}$*. To save some of the escaping heat, Verča plugged the hole with two bricks o cross-sections* $S_1 = a \times (b/3)$ *and* $S_2 = a \times (2b/3)$*, and of thermal conductivity coefficients* $\lambda_1 =$ $= 0.80\,\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$ *and* $\lambda_2 = 1.30\,\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$*, respectively. The lengths of both bricks are th same as the thickness of the wall* $s_1 = 15\,\mathrm{cm}$*, so they filled in the hole perfectly. Nevertheless t would be nicer to be able to look outside. What thermal conductivity coefficient* λ *would homogeneous glass panel of thickness* $s_2 = 3.0\,\mathrm{mm}$ *need to have if it conducted heat from th shack in the same way as the two bricks? The glass panel would be inset in the hole parallell to the wall (like a window). Assume that the rest of the shack insulates perfectly.*

Verča gets lost in other places than just lectures

Problem 17 ... moving pulleys on a wagon

5 point

We have a wagon in the shape of a cuboid, which is moving with an acceleration of $a = 1.2\,\mathrm{m{\cdot}s^{-2}}$. *On the top of the wagon, a smaller cuboid of mass* $m_1 = 15\,\mathrm{kg}$ *is placed, and it is connected by a rope over a pulley with another cuboid of mass* $m_2 = 10\,\mathrm{kg}$, *which is hanging on the rope in front of the wagon, touching its front wall. With what acceleration (with respect to the wagon) are the small cuboids going to move? The static friction coefficient between the small cuboids and the wagon is* $f = 0.15$. *Assume that the rope and pulley are massless.*



Legolas truly enjoys adjusting his own problems

Problem 18 ... beware of the fountain

4 point

There is a circular fountain at a town square with a diameter of $D = 6\,\mathrm{m}$. At its center, a sprin of water gushes vertically upwards to a height of $h = 4\,\mathrm{m}$. What is the lowest speed a shor gust of wind has to blow at so that the gushing water falls outside the fountain? Assume tha the wind blows horizontally and transfers 20 % of its speed to water drops.

Danka was observing children swimming in a fountain

Problem 19 ... Beryllgläser

5 point

We have two thin lenses of the same shape. The first one is made out of plastic with a refractiv
ndex of $n_\mathrm{p} = 1.67$ *and has an optical power of* $\varphi_\mathrm{p} = 4.20\,\mathrm{D}$*. The second lens is made out o*
eryl and has a refractive index of $n_\mathrm{b} = 1.57$*. What is the optical power of the second lens*
Both of the lenses are surrounded by air. *Karel hear*
hat eyeglasses were made out of beryl in old times, which inspired its German name, Brille

Problem 20 ... filling the bathtub

3 point

Robert of mass $m = 60\,\mathrm{kg}$ *and of the same density as water wanted to measure the volumetri flow rate of water in the tap while filling his bathtub. The bathtub has a rectangular base o dimensions* $150\,\mathrm{cm} \times 75\,\mathrm{cm}$. *Robert filled the bathtub with water to the height* $h = 20\,\mathrm{cm}$, *whe he had* $p = 80\,\%$ *of his body under the water. It took him* $t = 8\,\mathrm{min}$ *to fill the bath. Find th volumetric flow rate of water in the tap.* *Robert spends too much time in the bathtub*

Problem 21 ... Lipno and relativity

5 point

Jarda found out that running is physically demanding. He wants to accelerate himself to th speed of 99 % *of the speed of light in vacuum; however, he would need lots of energy. To imagin such an amount of energy, he calculated how many times he would have to heat and evaporat the entire Lipno reservoir, which has a volume of 310 million cubic meters. Assume the initia water temperature is always* 20 °C*. Jarda weighs* 75 kg*. If Jarda's calculations are correct, wha number did he get?*

Jarda ran out of energy

Problem 22 ... the intersections of Lissajous curves

5 point

There is a point that oscillates along the x axis according to the equation $x = x_{\mathrm{m}} \sin 5\omega t$ an along the y axis according to $y = y_{\mathrm{m}} \sin 6\omega t$, where x_{m} and y_{m} are amplitudes (which are no necessarily the same), t is time and 5ω and 6ω are angular frequencies of oscillations. How man intersection points of the motion graph are there in the xy plane?

Karel loves to create problems on Lissajous figures

Problem 23 ... conductor in a hurry

5 point

At the beginning of every rehearsal session of the symphonic orchestra, all the players tune thei nstruments. The process begins with a piano or an oboe giving off the tone A (in this problem ve assume it has a frequency $f_{\mathrm{A}} = 443\,\mathrm{Hz}$*), and the other musicians tune their instrument ccording to them. Imagine that the conductor is still running around during the process wit elocity* $v_{\mathrm{c}} = 3\,\mathrm{m{\cdot}s^{-1}}$*. Suppose that when running from the oboist to the pianist, the soun vaves of their instruments interfere. What beat frequency is he going to hear?*

Vojta was tuning his cello unsuccessfully

Problem 24 ... height does not matter

4 point

Consider a classical mathematical pendulum – a point mass on a massless string. Such a pendu
um is placed in a hot air baloon and is left to rise through the atmosphere. The temperature t o
he surrounding air decreases linearly with the height above the sea level h as $t = t_0 - kh$, wher
$_0 = 25\,^\circ\mathrm{C}$ *and* $k = 0.007\,\mathrm{K{\cdot}m^{-1}}$. *At the same time, however, the magnitude of the Earth'*
ravitational field slowly decreases as the height increases. We want the period of oscillation
f the pendulum to remain independent of the height h. What should the (non-zero) value o
he thermal expansion coefficient α of the string be?

Jarda gets cold on the 16th floor of dormitories

Problem 25 ... the hollow Earth has many variants

5 point

How much would the value of the gravitational acceleration at the Earth's surface decrease if ou planet was only a spherical shell of the same outer radius the Earth has now and of thickne $D = 100.0\,\mathrm{km}$? Consider the spherical shell to be homogeneous and of the same density a the Earth's average density. The result should be the ratio of this acceleration to the actua gravitational acceleration acting at the Earth's surface.

Karel keeps returning to the topic of spheres

Problem 26 ... an oscillating pulley

5 point

Consider a weightless pulley hanging from the ceiling on a spring with stiffness $k = 80\,\mathrm{N/m}$. A weightless rope is passed over the pulley. While one end of the rope is attached to the ground there is a body of mass $m = 1\,\mathrm{kg}$ attached to the other end. If we pull the body slightly downwards and then release it, what will the period of small oscillations of the system be?

Lego loves oscillations and pulleys, so he finally put them together

Problem 27 ... one increased, the other increased too

5 point

Lego was captured by cannibals. They imprisoned him in a hut, where he had an ideal DC volt
ge source, many perfect conductors and several adjustable resistors (rheostats). To stay alive
Lego must create a specific electric circuit. In such a circuit, there has to be at least one rheosta
uch that increasing its resistance causes the current flowing through some other rheostat (i
he same circuit) to increase. What is the minimum number of rheostats this circuit needs t
ontain? If such a circuit does not exist, the answer should be 0.

Lego has no clue whether anyone could fill themselves up by him

Problem 28 ... Doppler on his way home

6 point

Christian Doppler was on his way home. After a while, he noticed that people in both direction
(the same as his and the opposite) walk at a speed of $v = 1.0\,\mathrm{m{\cdot}s^{-1}}$ *and have distances of* $l =$
$= 4.0\,\mathrm{m}$ *between them. Doppler decided to take the chance to find the speed at which he shoul*
walk in order to meet as few people as possible on his way home. What is the minimum numbe
of people that Doppler will meet on his way home if he's $d = 5.2\,\mathrm{km}$ *away from his home? Yo*
may assume that $d \gg l$*, do not consider relativistic effects.* *Legolas is kind of autistic*

Problem 29 ... psoon

5 point

Jarda took a spoon out of a dishwasher and hung it on a cutlery stand. Obviously, Jarda doe not hang cutlery perfectly, so the spoon is now swinging from side to side. Determine the perio of its oscillations if we approximate its shape to be planar, composed of a circle of radius R = = 1.5 cm connected to a rectangle of width $a = 0.7\,\mathrm{cm}$ and length $b = 9\,\mathrm{cm}$. The spoon is 1 mm thick and is made of a material with density $\rho = 8000\,\mathrm{kg{\cdot}m^{-3}}$. The spoon is hanging by a sma hole which is located $s = 1\,\mathrm{cm}$ from the end of the spoon and in the middle of its width.

Jarda lacks a dishwasher in the dorm

Problem 30 ... on a beach

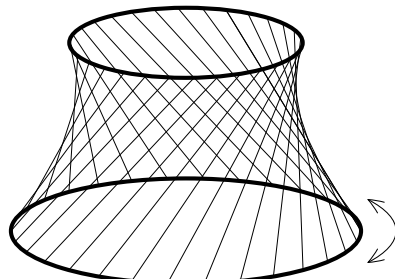
6 point

Jarda is playing ball games with his friends on the beach. In water, Jarda can move with a speed of $0.7\,\mathrm{m\cdot s^{-1}}$*, while on land, he can move with a speed of* $1.0\,\mathrm{m\cdot s^{-1}}$*. Jarda is standing exactly on the straight line which forms the border between water and land and knows that the ball will fall near him in exactly* 3 *seconds. What is the area of the region (on land or water) where the ball might fall in such a way that Jarda will be able to catch it before it hits either land or water?*

Jarda spent some time on a beach in Greece instead of creating problems

Problem 31 ... an oscillating hoop reloaded

6 point

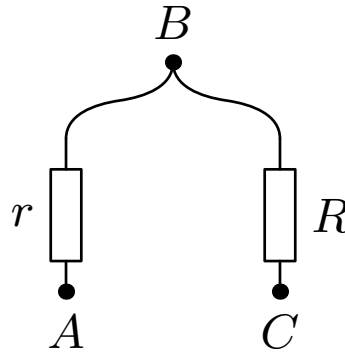


We have two hoops. The upper one has a radius $r = 1.0\,\mathrm{m}$ *and is fixed n a horizontal plane, and the bottom hoop has a radius* $R = 1.5\,\mathrm{m}$*. The ottom one is attached to the upper hoop by several massless cords, each f which has the same length, and if we look at the hoops from the top, we ee the cords pointing radially. Therefore, the bottom hoop "lies" in the orizontal plane, too. The distance between the plane of the upper hoop nd that of the bottom hoop is* $h_0 = 2.0\,\mathrm{m}$*. Let the mass of the bottom hoop be* $m = 1.0\,\mathrm{kg}$*. I ve rotate the bottom hoop a bit around its vertical axis and release it, what is the period o ts small oscillations?*

Lego adjusted his problem from last year

Problem 32 ... finite circuit

6 point



n the figure, you can see a fragment of an electrical circuit with resistors $r =$ $= 2.35\,\Omega$ *and* $R = 271.2\,\Omega$*. Any number of fragments can be connected ogether in a series according to several rules:*

- *Each connector A (except for the first fragment) must be connected to exactly one B.*
- *Similarly, each B (except for the last fragment) must be connected to one A.*
- *All connectors C must be connected at one point.*
- *No other connections are allowed.*

Then we connect the A and C of the first fragment with one terminal of multimeter, and we connect the B of the last fragment to the second erminal. What is the minimum number of fragments we need to use in rder to measure a resistance greater than $R_x = 23.7\,\Omega$*?*

Jáchym wanted to come up with a problem about springs, but somehow it didn't work ou

Problem 33 ... an inclined roof

6 point

t is raining so much that 20 mm *of precipitation falls in an hour. The rectangular roof o n older building with dimensions* $a = 20\,\mathrm{m}$ *and* $b = 10\,\mathrm{m}$ *has a slope which is very small, bu ufficient for all the water from the roof to flow into one place and fall to the ground fron here. Determine the force which the stream of falling water exerts on the ground if it bounce pwards after impact with speed* $k = 5$*-times smaller than it was just before impact. Wate drains from the roof so fast that a constant amount of water is maintained on it. The roof is a height of* $h = 3\,\mathrm{m}$ *above the ground. Assume that the water draining from the roof has zer nitial vertical velocity.*

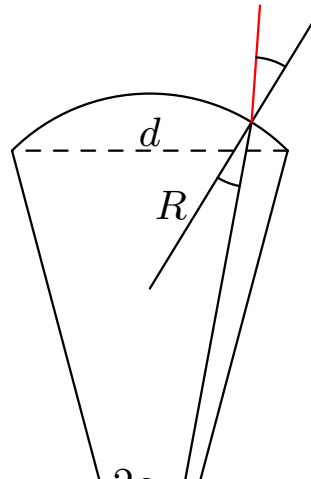
Danka was walking in the rain around a building with an inclined roo

Problem 34 ... ommatidium

6 point

The eyes of insects are composed of many individual small units, the so-called ommatidia. We can consider each ommatidium to be a solid cone with vertex angle $2\alpha_0 = 14°$. While photoreceptors are located at the vertex, the other side of the cone is enclosed by a spherical surface of radius $R = 17\,\mu\text{m}$, which is coaxial with the cone. The diameter of the base of the cone is $d = 25\,\mu\text{m}$. Determine the solid angle that one ommatidium can perceive if its interior is filled with a material of refractive index $n = 1.3$.

Jarda tried a school student project.



Problem 35 ... thermodynamic cannon

6 point

We have a horizontally placed cylinder with a base area $S = 2.5\,\mathrm{dm}^2$ and a length $l = 2\,\mathrm{m}$ One end of the cylinder is closed and the other end is not. In the cylinder, there is a pisto with a weight of $m = 24\,\mathrm{kg}$, which can move without friction and seals perfectly. The volume o he gas between the piston and the closed end is $V_0 = 3.5\,\ell$. First, everything is in equilibriu – the pressure outside the cylinder and also between the piston and the closed end is equa o normal atmospheric pressure and the temperature is T_0 everywhere. However, suddenly a xplosion occurs – the temperature of the enclosed part of the gas changes to $10T_0$, and th mount of substance there becomes twenty times greater. At what speed does the piston leav he cylinder? Assume that classical equilibrium thermodynamics apply and the heat capacit atio of the gas is $\kappa = 1.4$.

Lego wanted to play a joke on someone

Problem 36 ... center of gravity of a snail

6 point

How far away from the center of a spiral is its center of gravity? Consider a spiral with constan linear density λ, given in polar coordinates as $r = a\varphi$, where $a = 0.1\,\mathrm{m{\cdot}rad^{-1}}$ and φ is the pola angle in radians. The total angle subtended by the spiral is 10π.

Jarda fights snails in his flowerbed

Problem 37 ... very thirsty

6 point

While wandering through a desert, Jarda came across a canister of water. The canister ha a cuboid shape. It has width $c = 20\,\mathrm{cm}$, *length* $b = 40\,\mathrm{cm}$ *and height* $a = 60\,\mathrm{cm}$. *The caniste does not have a lid, but it has a tap through which water can flow out, in the middle of on of the bottom edges of length c. The container is 90 percent full. Jarda is so thirsty that h wants the water to flow from the tap as fast as possible. What is the angle (with respect to th vertical) by which he needs to tilt the canister around the edge c? Jarda first tilts the caniste and then opens the tap.*

Jarda was working at the steppes of South Moravia

Problem 38 ... rotary capacitor

6 point

A capacitor consists of two rotary semicircular plates with the same radii $R = 2\,\mathrm{cm}$ *at a mutua distance of* $d = 0.1\,\mathrm{cm}$, *separated by air. In the initial state, the plates are exactly on top o each other, the capacitor is charged to a voltage* $U_0 = 20\,\mathrm{V}$ *and then the voltage source i*

disconnected. When rotating one plate away from the other (around the axis passing throug
he midpoint of the straight side of both plates), we act with a torque. What is the initia
magnitude of this torque?

Jarda couldn't loosen a screw

Problem 39 ... inevitable fall

7 point

Imagine a body of a weight $M = 72\,\mathrm{kg}$, *which we release from a height of* $h = 12\,\mathrm{m}$ *abov the ground. At that moment, we start firing at it with a machine gun with a rate of fire* $f = 2\,000\,\mathrm{min}^{-1}$ *from the point where it would soon fall (the first projectile is shot at* $1/f$ *afte releasing). The projectiles have a mass of* $m = 11.3\,\mathrm{g}$, *move at a speed of* $v = 790\,\mathrm{m{\cdot}s}^{-1}$ *an after collisions, they stay inside the body. How long would it take for the body to fall to th ground?* *Jáchym knew it would eventually fall, but he wanted to know when exactly*

Problem 40 ... life is short

7 point

Matěj found a piece of pure radium 225. After a while, he discovered that it decayed by β^- decay to actinium with a half-life of 15 days. However, the story does not end here. Actinium is subject to α decay to francium with a half-life of 10 days. In how many days (from the moment he found the radium) would Matěj have had the maximum amount of actinium?

Matěj stole this from a lecture on Computer Methods

Problem 41 ... radiant

5 point

An ideal black body radiates with the maximum intensity I_λ at a wavelength of $\lambda_{\max} = 598.34 \cdot 10^{-9}$ m. At what wavelength does the black body radiate with half of the maximum intensity? Submit the largest correct answer.

Danka thought she wouldn't like to solve this problem

Problem 42 ... anti-reflective glasses

7 point

Patrik has a thin anti-reflective layer on his glasses of a thickness $d = 250\,\mathrm{nm}$ *and of a refractiv index* $n_2 = 1.4$. *Which part of the visible light is subject to destructive interference? Th efractive index of air is* $n_1 = 1$, *for glass it is* $n_3 = 1.55$ *and light rays are perpendicular to th lasses.*

Patrik was about to buy sunglasses

Problem 43 ... Dano is flying to an exoplanet

7 point

For exoplanet research, Dano would like to fly to Proxima Centauri, at a distance of 4.2 light years. Calculate the Lorentz factor if Dano wants to get exactly half a year older on his way there and back. Do not consider acceleration of the spaceship.

Vašek was jealous of Dano's research on exoplanets

Problem 44 ... platform 9 and 3/4

6 point

We are standing on a train platform 2 m *away from the track. Next to us, there is a very lon*
rain which is slowly braking. We are located roughly near the middle of the train. We hea
really unpleasant squeal of brakes with loudness of 110 dB *coming from the entirety of th*
rain. Moving away from the track might help us a bit. How many decibels do we measure a
he other side of the platform, 7 m *away from the track?*

Jarda believes in Harry Potter and loves Hermione

Problem 45 ... a ride on the exponential

8 point

The exponential is an essential function in physics, which is why Jarda built a roller coaste
n the shape of this function in his new amusement park. The ride starts at a height of h =
= 10 m. The cart then goes down a path which, if viewed from the side, has a profile give
by the function $y = he^{-\frac{x}{h}}$*, where* x *is the distance in the horizontal direction from the bas*
f the entrance tower at the beginning of the ride. At what height above the ground does th
descending cart exert the greatest force on the roller coaster?

Jarda is tired of taking the elevato

Problem 46 ... carousel ride

7 point

n addition to the roller coaster, Jarda also bought a chain carousel for his new amusemen
ark. A horizontal disc with a diameter of $d = 8\,\mathrm{m}$ *rotates with a period of* $T = 8\,\mathrm{s}$*, and th*
eats are attached to the edge of the disc using ropes of length $l = 8\,\mathrm{m}$*. Determine the forc*
xerted by a child weighing $m = 25\,\mathrm{kg}$ *on a seat when the carousel is in a steady state.*

Jarda works around the clock

Problem 47 ... catapult

7 point

We thread a small bead on a rod of a length $l = 30\,\mathrm{cm}$, which lies on a horizontal pad and ca
nly rotate vertically around one of its ends, which is fixed on the pad. We start to rotate th
od upwards at a constant angular speed. What is the maximum distance from the center o
otation at which we can place the bead such that it leaves the rod just at the moment whe
t reaches the vertical position? The bead moves along the rod without friction.

Jarda wanted to hit the professor while in a lecture on Theoretical Mechanics

Problem 48 ... too many cubes

5 point

Consider an infinite cube net consisting of identical resistors $R = 24\,\Omega$, which are located o
the edges of all cubes. What resistance do we measure between two adjacent vertices?

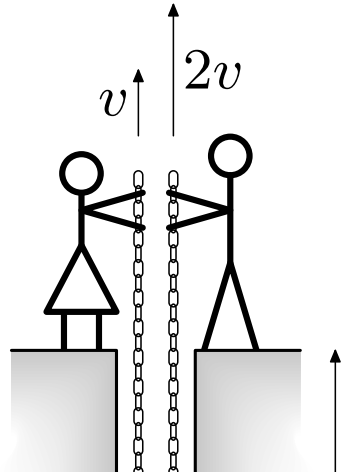
Jarda wanted a challenging problem with a short solution

Problem 49 ... we work by pulling

8 point

Alice and Bob are pulling a non-elastic chain of linear density $\lambda = 0.4\,\mathrm{kg\cdot m^{-1}}$ *from a pit. Alice is pulling the chain out at a constant speed* $v = 0.6\,\mathrm{m\cdot s^{-1}}$ *and Bob at twice the speed. Consider the work done between the point in time when the lowest point of the chain is at a depth of* $L = 5\,\mathrm{m}$ *and when it is fully pulled out. How many times larger will Bob's work be than Alice's? Assume that they have already been pulling for some time at the starting point.*

Matěj is fascinated by the Mould effect.



Problem 50 ... snowball to a window

8 point

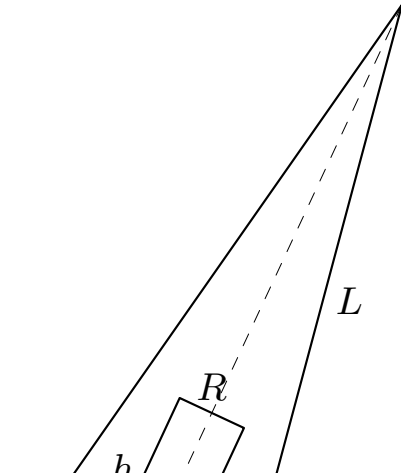
Jarda wanted to distract his quarantined friend, so he decided to throw a snowball at hi window. The lower frame of the window is at a height of $h_1 = 3.5\,\mathrm{m}$ *and the upper fram at* $h_2 = 4.8\,\mathrm{m}$*. The window is* $d = 2\,\mathrm{m}$ *wide. Jarda was standing four meters from the base o the house, right in front of the center of the window. What part of the area of the window (i percent) could he hit if he threw the ball at a speed of* $v = 8\,\mathrm{m{\cdot}s^{-1}}$*, from* $H = 1.8\,\mathrm{m}$ *abov ground?*

Jarda aims high

Problem 51 ... a bottle on a swing

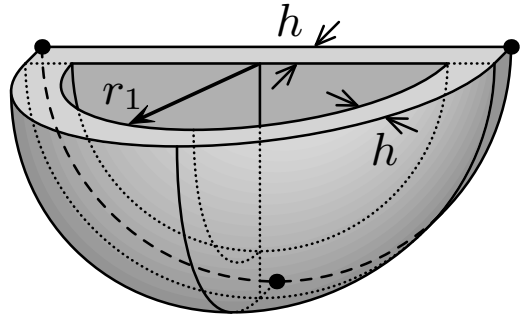
8 point

Jarda is swinging harmoniously on a swing with a frequency $f = 0.3\,\mathrm{Hz}$. *He has a full cylinder-shaped bottle with him, placed exactly in the middle of the width of the swing. The bottle has a height* $h = 30\,\mathrm{cm}$ *and it stands on its base of a diameter* $R = 3.0\,\mathrm{cm}$. *The swing is* $D = 40\,\mathrm{cm}$ *wide and hangs on ropes that are* $L = 1.4\,\mathrm{m}$ *long. The friction between the swing and the bottle is enormous, so the bottle stands in its place. Determine the smallest amplitude of the swing's inclination angle upon which the bottle starts to wobble.* *Jarda keeps dropping things.*



Problem 52 ... washbasin

8 point



A washbasin has two parts: the front part is shaped like a quarter of a sphere, with an inner radius of $r_1 = 30\,\mathrm{cm}$*, and it is attached to the back part in the shape of a half-cylinder. The thickness of all its walls is* $h = 2.0\,\mathrm{cm}$ *and the whole washbasin is made of a material with a density of* $\rho = 3\,100\,\mathrm{kg{\cdot}m^{-3}}$*. It is attached to a wall at two points located at its only two vertices (corners). Other than that, its bottom edge leans freely on the wall. What is the minimum force which the hanging mechanism at each corner must withstand when the basin is completely filled with water? Assume that the bottom edge acts on the wall only at its lowest point and only in the direction perpendicular to the wall.*

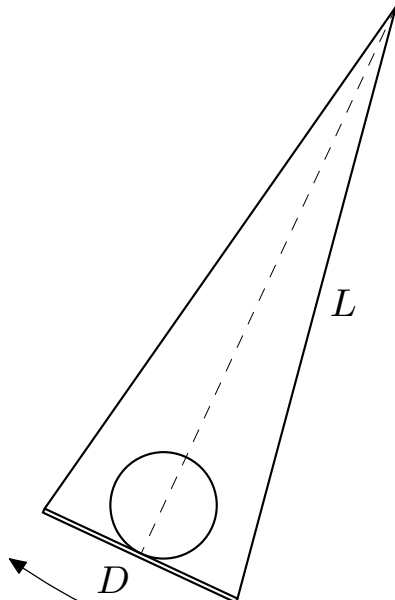
Dodo was repairing a bathroom tap

Problem 53 ... bottle on a swing reloaded

8 point

Jarda is swinging harmoniously on a swing with a frequency $f = 0.5\,\mathrm{Hz}$ *and a maximum inclination angle* $5°$*. Next to him, there is an empty, relatively narrow, and long cylinder-shaped bottle, exactly in the middle of the width of the swing. The bottle is placed horizontally, and its axis is perpendicular to the direction of motion. The swing is* $D = 40\,\mathrm{cm}$ *wide and hangs on ropes that are* $L = 3\,\mathrm{m}$ *long. At first, Jarda is holding the bottle, but he releases it when he passes through the equilibrium position. When does the bottle fall off the swing? Assume that the bottle is rolling without slipping.*

Jarda drank a bottle of wine and started thinking about problems.



Problem 54 ... minimisation of acceleration

7 point

We are driving a car, which we consider to be a point mass, at a speed of $v_0 = 20.0\,\mathrm{m\cdot s^{-1}}$*. A ome point, we notice that we are approaching a curve, at the end of which we'll need to sto t traffic lights. Due to our insurance company, we want the maximum absolute value of th cceleration acting on the car (until it stops at the traffic lights) to be as small as possible. If w tart braking* $d = 30.0\,\mathrm{m}$ *before the beginning of the curve, the radius of the curve is* $R = 10.0\,\mathrm{m}$ *nd we turn* $22.5°$ *on the curve, what time does it take to reach the end of the curve from th oint where we start braking? Karel's insurance company insists on minimum acceleration*

Problem M.1 ... the boat

3 point

Lego wants to cross a river of a width $l = 5.0\,\mathrm{m}$ *on a small boat. The maximum speed at whic*
he can paddle is $v_\mathrm{B} = 1.0\,\mathrm{m{\cdot}s^{-1}}$ *and water flows at speed* $v_\mathrm{F} = 0.50\,\mathrm{m{\cdot}s^{-1}}$.

What is the fastest time in which Lego can cross the river if he does not care how far alon
he shore the current takes him? *Lego just got carried away*

Problem M.2 ... the boat reloaded

3 point

Lego wants to cross a river of a width $l = 5.0\,\mathrm{m}$ *on a small boat. The maximum speed at whic*
he can paddle is $v_\mathrm{B} = 1.0\,\mathrm{m{\cdot}s^{-1}}$ *and water flows at speed* $v_\mathrm{F} = 0.50\,\mathrm{m{\cdot}s^{-1}}$ *in the whole river.*

What is the fastest time in which Lego can cross the river if he wants to move in the directio
perpendicular to the water flow at all times?

Lego found out that the shortest path is not always the fastest one

Problem M.3 ... the boat revolutions

3 point

Lego wants to cross a river of a width $l = 5.0\,\mathrm{m}$ on a small boat. The maximum speed at whic he can paddle is $v_\mathrm{B} = 1.0\,\mathrm{m{\cdot}s^{-1}}$. He wants to move in the direction perpendicular to the rive again; however, it has been raining and the water seems to flow faster. What is the slowes speed of the flow for which he cannot cross the river?

Legolas needed one more problem for the series

Problem M.4 ... the boat resurrections

4 point

Lego wants to cross a river of a width $l = 6.0\,\mathrm{m}$ on a small boat. The maximum speed at whic he can paddle is $v_\mathrm{B} = 1.0\,\mathrm{m{\cdot}s^{-1}}$. The water flows at a speed $v_\mathrm{F} = 1.5\,\mathrm{m{\cdot}s^{-1}}$. From the previou problem, Lego already knows that in this situation, it is impossible to move in the directio perpendicular to the river. He would like to know the minimum distance by which the curren would take him away (from the point on the other shore of the river directly opposite to hi starting point) while crossing the river. Lego optimizes

Problem E.1 ... discharging a capacitor

3 point

We are discharging a capacitor of a capacitance $C = 42.0\,\mathrm{mF}$ *through a resistor with a resistanc* *of* $R = 9.81\,\mathrm{k\Omega}$. *We know that the equation* $u(t) = U_0 \mathrm{e}^{-\frac{t}{RC}}$ *holds for the evolution of voltag* *n time. In what time will the capacitor's electric field have half the energy of the fully charge* *tate?* *Karel wanted the participants to get familiar with discharging*

Problem E.2 ... conductive chewing gum

4 point

Rado is chewing a chewing gum with a specific electrical resistance $\rho = 1\,\mathrm{m\Omega{\cdot}mm}$ *and volum* $V = 1\,\mathrm{cm}^3$. *He splits it in half and uses each half as a separate conductor. In his backpack he has an electrical appliance of an internal resistance* $R = 1\,\Omega$ *that can only be powered by voltages up to* $U_{\max} = 5\,\mathrm{V}$, *and a DC voltage source* $U = 12\,\mathrm{V}$. *What is the minimum distanc between the electrical appliance and the power supply if he wants to connect them with th two conductors made of chewing gum? The conductors should have the same constant cross ections.*

Rado had chewing gum after a long time

Problem E.3 ... cheap voltmeter

3 point

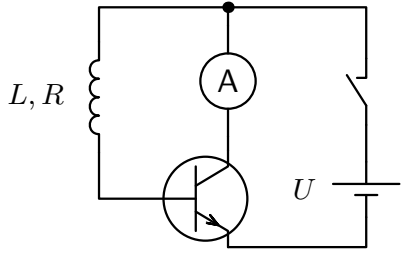
An AC voltmeter works as follows. At first, the DC component is filtered out of the inpu voltage using a high-pass filter. Subsequently, the obtained voltage is rectified and smoothed b a low-pass filter. The resulting DC voltage is measured and converted to the displayed value b multiplying it by a numerical factor. This factor is chosen in such a way that for a sine wave the displayed value is the effective value of voltage. What value is being displayed if the inpu AC voltage has a sawtooth waveform oscillating between $U_1 = 0.00\,\mathrm{V}$ *and* $U_2 = 1.00\,\mathrm{V}$*?*

Dodo steals ideas for problems from the Department of Physics Education

Problem E.4 ... coil on a base

4 point

Consider a circuit with a DC source ($U > 50\,\mathrm{V}$), an inductor with inductance $L = 5\,\mathrm{H}$ and resistance $R = 10\,\mathrm{k\Omega}$, an NPN transistor, an ideal ammeter, and a switch. Some time after closing the switch, the value on the ammeter will settle on $I = 1\,\mathrm{A}$. At what time after closing the switch does the ammeter display the value $I' = 500\,\mathrm{mA}$?



Vojta was not building a bomb.

Problem X.1 ... I'll pull her down with me

3 point

Let us assume we have a ball of air of radius $R = 2.1\,\mathrm{cm}$, *which is wrapped in a layer o polystyrene of thickness* $t = 2.1\,\mathrm{cm}$ *and density* $\rho = 33\,\mathrm{kg{\cdot}m^{-3}}$. *We attach the ball to a force gauge by a rigid, weightless string and then pull it down so that half of the ball's volume is under the water surface. What is the value we see on the force gauge?*

Karel had balloons on his mind

Problem X.2 ... to the bottom of the dam

4 point

Consider a piece of polystyrene with weight $m_0 = 1.25\,\mathrm{kg}$ *and density* $\rho_0 = 27\,\mathrm{kg{\cdot}m^{-3}}$ *tha*
we want to completely submerge to the bottom of a water dam. We have plenty of stones o
various sizes and the same density $\rho = 2\,650\,\mathrm{kg{\cdot}m^{-3}}$. *What is the minimum weight of a ston*
which we should attach to the polystyrene? *Karel was thinking about Archimedes*

Problem X.3 ... an unknown ball in an unknown fluid

4 point

We are given a ball of unknown radius, submerged in a fluid of unknown (but constant) density
We know that the difference between hydrostatic pressures at the top and the bottom point o
he ball is $\Delta p = 850\,\mathrm{Pa}$*, and the total buoyant force acting on the ball is* $F_\mathrm{b} = 150\,\mathrm{N}$*. What i*
he density of the fluid?

Lego was inventing buoyancy problems

Problem X.4 ... attractive

3 point

Danka has two brass balls with diameters of $2.00\,\mathrm{cm}$ *carrying the same charges. One of the ball*
s fixed and the other one hovers above it at a distance of $d_1 = 20.0\,\mathrm{cm}$ *(we assume that the ba*
an only move in the vertical direction). We now immerse both of the balls in oil of density $\rho_0 =$
$= 910\,\mathrm{kg{\cdot}m^{-3}}$*, where a new equilibrium is established. The balls are now* $d_2 = 14.0\,\mathrm{cm}$ *awa*
rom each other. What is the relative permittivity of the oil we used? Consider the densit
f brass to be $\rho = 8\,400\,\mathrm{kg{\cdot}m^{-3}}$*. Distance between the two balls means the distance of thei*
enters.

Danka was playing with balls